System Architecture Notes

TechNova Solutions

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1. Purpose

The purpose of this document is to describe the high-level system architecture of TechNova Solutions and identify the primary technical components, data flows, access paths, and security-relevant dependencies within the environment.

This document supports the ISMS scope, asset inventory, risk assessment, and control mapping activities.

- Company: TechNova Solutions
- Industry: SaaS startup
- Environment: web app, backend API, production database, cloud hosting, employee laptops, admin access, logging, backups
- Team size: 10–20 employees
- Main concern: protection of customer data and continuity of service

2. Environment Overview

TechNova Solutions operates a cloud-hosted SaaS platform accessed by customers through a web browser. The platform is supported by backend application services, a production database, administrative interfaces, logging mechanisms, and backup storage.

The company has a small team of approximately 10 to 20 employees. Some personnel require access to internal systems or administrative interfaces in order to support operations, maintain infrastructure, or assist customers.

The architecture is lightweight and startup-oriented but includes business-critical assets that require structured security oversight.

3. High-Level Architecture Components

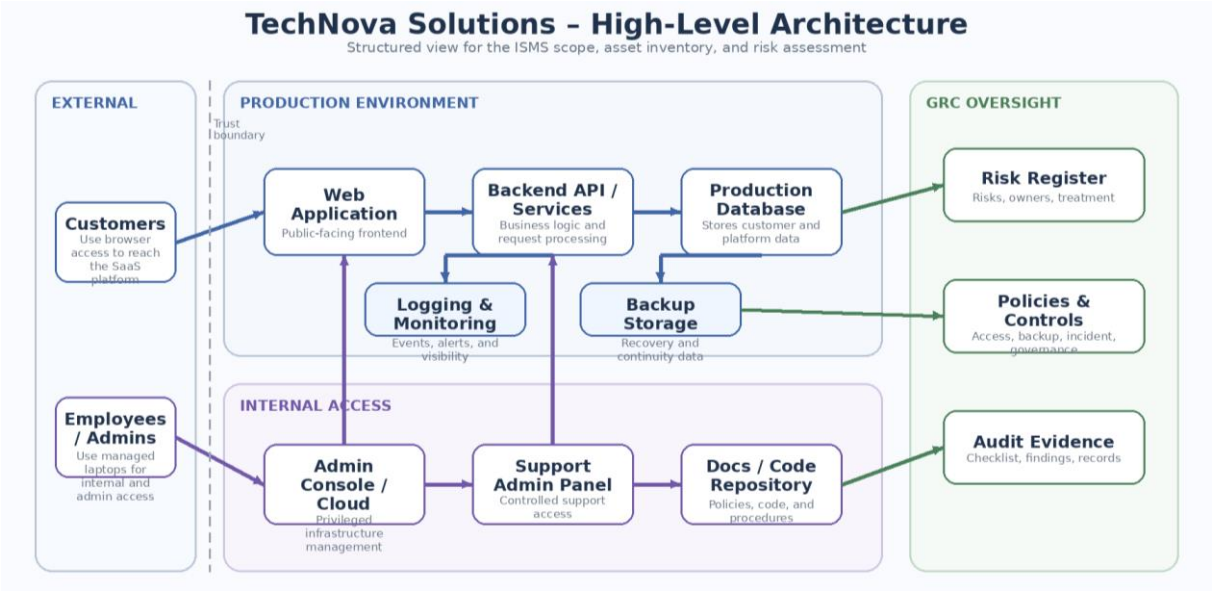
The TechNova environment includes the following main components:

3.1 Customer-Facing Web Application

The web application is the primary interface through which customers access TechNova's SaaS service. It is reachable over the internet and serves as the entry point for customer authentication, account access, and platform usage.

3.2 Backend Application Services

Backend services process business logic, receive requests from the web application, interact with the database, and handle internal application functionality.



3.3 Production Database

The production database stores customer records and core application data. It is one of the most sensitive components in the environment because it contains business-critical and customer-related information.

3.4 Cloud Infrastructure

TechNova's application and supporting services are hosted within a cloud environment. This includes compute, storage, administrative management interfaces, and platform-level configuration.

3.5 Administrative Accounts and Consoles

Privileged users access administrative consoles to manage infrastructure, services, users, configurations, and operational settings. These accounts represent a high-value security target.

3.6 Logging and Monitoring

The environment includes logging and monitoring capabilities used for operational troubleshooting, event visibility, and incident detection.

3.7 Backup and Recovery Storage

Backups are maintained to support business continuity and recovery in the event of service disruption, data corruption, or security incidents.

3.8 Employee Endpoints

Authorized employees use company-issued laptops to access internal resources, cloud consoles, support interfaces, documentation repositories, and business systems.

4. Main User Types

The following user groups interact with the environment:

4.1 Customers

Customers access the TechNova SaaS platform through the public web application. Their interactions may include authentication, viewing data, entering records, and using platform features.

4.2 Engineering Personnel

Engineering personnel manage application code, deployments, debugging, and service configuration. They may access development tools, repositories, and restricted environments.

4.3 Operations / Infrastructure Personnel

Operations personnel manage infrastructure, backups, monitoring, and service availability. Their access may include cloud consoles and operational tooling.

4.4 Customer Support Personnel

Customer support personnel may access limited customer information or support panels in order to assist users and resolve issues.

4.5 Management

Management may access reports, dashboards, internal documentation, and risk/compliance records.

5. Core Data Flows

The main data flows in the TechNova environment are as follows:

5.1 Customer to Web Application

Customers connect to the web application over the internet and submit login credentials, account requests, and application data.

5.2 Web Application to Backend Services

The frontend communicates with backend services to process requests, enforce business logic, and retrieve or update information.

5.3 Backend Services to Production Database

Backend services store and retrieve customer and application data in the production database.

5.4 Admin Users to Administrative Consoles

Authorized internal users connect to cloud administration consoles, support panels, repositories, or internal management interfaces.

5.5 Systems to Logging and Monitoring

Application, infrastructure, and access events generate logs that support system visibility, troubleshooting, and security monitoring.

5.6 Production Environment to Backup Storage

Backup processes copy selected data and system content to protected storage for recovery purposes.

6. Security-Relevant Trust Boundaries

Several trust boundaries exist within the architecture and are important for risk analysis:

6.1 Internet to Public Application Boundary

The web application is publicly reachable and therefore exposed to internet-based threats such as unauthorized access attempts, application attacks, and denial-of-service activity.

6.2 Application to Database Boundary

The communication path between application services and the production database is highly sensitive because compromise at this boundary may affect customer data confidentiality and integrity.

6.3 User Endpoint to Administrative Access Boundary

Employee laptops are used to access sensitive systems. If endpoints are compromised, attacker access may extend into privileged environments.

6.4 Administrative Access Boundary

Privileged accounts used in cloud or system administration represent a critical trust boundary due to the level of access they provide.

6.5 Backup and Recovery Boundary

Backup repositories are critical to recovery and may also contain sensitive replicated data.

7. Key Security Dependencies

TechNova's secure operation depends on the following capabilities:

- secure authentication and access control,
- appropriate privilege management,
- secure handling of customer data,
- visibility into logs and events,
- resilience through backup and recovery,
- endpoint protection for employee devices,
- and documented security procedures.

Weakness in any of these areas could materially affect confidentiality, integrity, availability, or incident response readiness.

8. Initial Security Observations

Based on the architecture and startup context, the following areas are likely to represent important security priorities:

- privileged access management,
- multi-factor authentication for administrative accounts,
- secure configuration of cloud resources,
- web application security,
- endpoint protection for employee laptops,
- backup testing,
- and sufficient logging and monitoring coverage.

These observations will be evaluated more formally during the risk assessment phase.

9. Architecture Summary

The TechNova Solutions environment is a cloud-based SaaS architecture composed of a public-facing web application, backend application services, a production database, administrative management interfaces, logging and monitoring components, and backup storage.

Because the company stores customer data and relies on a small team with limited operational complexity, the architecture is suitable for a lightweight but structured compliance program focused on high-priority security risks and foundational governance controls.